

MUHAMMAD BILAL

AI Full Stack Developer | MERN, LLM Integration & Agentic Systems

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C2C Available via iBuildsoft LLC

SUMMARY

Full Stack Engineer with 8+ years building production web applications on the MERN stack (MongoDB, Express.js, React, Node.js), delivered for US clients across healthcare, cybersecurity compliance, SaaS, and education, including current MERN-based delivery on an active compliance platform. Extending core full-stack expertise into AI-powered features: LLM integration, prompt engineering, and function/tool calling, with hands-on delivery of AI-driven analysis and voice AI capabilities in production systems. Experienced in HIPAA and CMMC/NIST environments. Available for Corp-to-Corp (C2C) contract engagements.

TECHNICAL SKILLS

MERN & Full Stack: React, Node.js, Express.js, MongoDB, Next.js, Vue.js, NestJS, TypeScript, JavaScript, PHP/Laravel, REST APIs, Microservices

AI & LLM Engineering: OpenAI API, LLM Integration, Prompt Engineering, Function/Tool Calling, Retrieval-Augmented Generation (RAG) fundamentals, Claude Code

Databases: MongoDB, PostgreSQL, MySQL, Redis, PouchDB, CouchDB

Cloud & DevOps: AWS (EC2, S3, Lambda), Firebase, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions

Compliance: HIPAA, NIST SP 800-171, CMMC

EXPERIENCE

iBuildsoft LLC

United States (Remote) Jan 2025 - Present

Full Stack Engineer

Project 1: Cybersecurity Compliance Platform (Current) - 2025 - Present

- Architectural assessment workflows, scoring logic, evidence tracking, and automated compliance evaluations aligned with CMMC and NIST SP 800-171 frameworks, using React, Node.js, Express.js, and MongoDB.
- Used MongoDB to store compliance evidence and AI-generated assessment output, whose structure varies by framework and control, avoiding constant schema migrations; used PostgreSQL for structured user, organization, and audit-trail data requiring relational integrity for reporting.
- Designed prompt templates and structured JSON output parsing for OpenAI-based compliance analysis, translating raw control language and uploaded evidence into scored gap findings and remediation recommendations.
- Built validation checks on LLM outputs (schema conformance, confidence flags for low-certainty mappings) to catch incomplete or unreliable compliance findings before they reached human reviewers.
- Automated gap detection, compliance recommendations, and audit reporting through this LLM pipeline, cutting manual review cycles significantly and reducing reviewer turnaround time.
- Provisioned and maintained AWS infrastructure (EC2, S3, Lambda) and CI/CD pipelines, improving deployment reliability while enforcing secure data handling and access controls across all services.
- Tuned database queries and indexing strategy to keep evaluation performance stable as compliance workloads and evidence volume grew.

Project 2: School Management SaaS - School Chain, Texas - 2025

- Developed LMS, finance, and cafeteria management modules for a multi-branch school platform using React, Node.js, Express.js, and MongoDB.
- Used MongoDB to track per-user video watch progress in the LMS module (resume position, completion state, quiz attempts), and PostgreSQL for structured student, billing, and branch records requiring relational integrity.
- Unified academic, financial, and identity data across branches by integrating Google Classroom, Clever, and QuickBooks, eliminating manual reconciliation between systems.
- Designed role-based dashboards and reporting tools for administrators, teachers, and finance staff, giving each role visibility into only the data relevant to them.

Confiz

Bellevue, WA (Remote) Sep 2023 - Dec 2024

Full Stack Engineer

Project 1: Chain Education Marketing Platform - 2023 - 2024

- Led React development for a legacy marketing platform serving a chain education brand, rebuilding core UX and adding new functionality.

- Integrated instructional video delivery, interactive quizzes, and instructor progress tracking, reducing admin workload by 2+ hours/week per instructor.
- Built real-time tracking and reporting dashboards using Firebase Realtime Database and Firestore, improving data processing response time by 40%.
- Here I implemented a mechanism that staff training should be tracked and stats should be aligned with the assigned video/ audio lectures.

Project 2: RAIN Healthcare App (HIPAA-Compliant) - 2024

- Resolved a critical timer desynchronization issue in an OR scheduling application deployed at Adam Vital Hospital by redesigning the event-driven architecture, achieving 100% clock alignment across all devices and timezones.
- Refactored the timer solution into reusable shared libraries later adopted by the mobile team, propagating the fix across all client platforms simultaneously.
- Integrated Speech-to-Text on the mobile side to power an AI-driven clinical voice response tool, letting hospital staff log updates and query OR status hands-free during procedures.
- Maintained strict HIPAA compliance throughout development; coordinated testing across hospital deployment environments.

LogLeads

Sydney, Australia Jun 2022 - Sep 2023

Frontend Engineer

- Led the migration of a Vue 2 SaaS lead generation platform to Vue 3, preserving full feature parity across the application.
- Shipped new real-time data transmission features using WebRTC to support live data exchange within the platform.
- Improved overall user experience through refined UI interactions and frontend performance optimizations.

10Pearls

Lahore, Pakistan Dec 2021 - Jun 2022

Senior Software Engineer

- Architected a Node.js, Redis, MongoDB backend achieving 99.9% uptime and 45% faster load speeds, integrated with a React/Next.js/Tailwind CSS frontend.
- Optimized server-side infrastructure to support 1M+ concurrent users with 60% latency reduction and zero downtime.
- Shipped a workflow automation UI serving 100K+ users, improving task completion efficiency by 35%.

Nextbridge Ltd.

Lahore, Pakistan Apr 2019 - Nov 2021

Full Stack Engineer

- Delivered full-stack applications for US clients across education, hotel management, and car rental industries in an Agile environment.
- Engineered an offline-first field operations tool with a web-based mobile screen designer: operators configured the mobile UI via web portal; data synced bidirectionally via PouchDB/CouchDB once devices were back online. Stack: Next.js, Node.js, Express.
- Developed microservices-based applications using the MERN stack (MongoDB, Express.js, React, Node.js) and MEAN stack; containerized services with Docker and managed live deployments.
- Refactored 5,000+ lines of legacy code, improving system performance by 35% and cutting release cycles by 2 weeks.
- Launched a digital library, podcast platform, and Alexa skill for US and Australian clients.

Raaziq International PVT LTD

Lahore, Pakistan Jan 2018 - Apr 2019

Software Engineer

- Engineered a Metro Bus management system and logistics tracking platform for government clients, automating workflows and eliminating 80+ hours of manual effort per month.
- Developed a multi-module SaaS ERP covering CRM, HRM, freight management, payroll, and inventory for 1,000+ users.
- Reduced SQL query execution time by 75% through targeted optimization; maintained 95% server uptime.

Wali Systems Inc.

Lahore, Pakistan Sep 2016 - Dec 2017

Software Engineer

- Built full-stack CRM and lead generation tools using Angular, Vue.js, Laravel, and Node.js; extended SugarCRM and SuiteCRM with custom modules and packages.
- Optimized client-side performance, reducing transaction cycle time by 2+ seconds per operation.
- Improved overall application reliability to 99.9% through systematic testing and debugging.

EDUCATION

University of Management and Technology (UMT), Lahore, Pakistan - Bachelor of Science in Software Engineering - 2012 - 2016

CERTIFICATIONS

NestJS Modern TypeScript Backend Development | React.js Essential Training - LinkedIn Learning | Problem Solving (Basic) Certificate - HackerRank | IEEE Programming Competition Participant

WORK AUTHORIZATION

Open to Corp-to-Corp (C2C) contract engagements via iBuildsoft LLC. Details available upon request.